Biraj Shrestha

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SUMMARY

Software developer with a B.S. in Computer Science and a 3.72 GPA, experienced in developing and maintaining production applications, integrating REST APIs, optimizing database performance, and automating workflows. Currently working at CloudRx on healthcare software and previously developed Banner ERP applications at Youngstown State University. Skilled in C++, PHP, JavaScript, Python, Java, SQL, and AWS, with additional experience building AI-powered tools using the Claude API and MCP.

PROFESSIONAL EXPERIENCE

CloudRx Pharmacy Hub — Farmers Branch, TX
Junior Software Developer — 02/2026 – Present

Integrations and data flows

- Build and maintain REST API integrations with enterprise EMR platforms, including Pioneer and Kindbody, that sync patient and clinical data between systems in a HIPAA-related environment and keep downstream data consistent.
- Integrated Mastercard Transaction Link Identifier (TLID) with the Chase Orbital payment gateway for end-to-end transaction lifecycle tracking and compliance on recurring billing and customer-initiated transactions.
- Designed an image retrieval workflow for Pioneer EMR (via its API) on AWS: the application checks the database first and, if no image is stored, calls the Pioneer API, saves the TIFF to Amazon S3, and stores the S3 path in the database. Later requests load directly from S3.
- Because a TIFF-to-JPEG converter could not be installed on the server, built a client-side JavaScript class that decodes TIFF bytes (custom LZW decompression using DataView) for browser display, with fallback handling when no image is available and a matching image view in the portal.

Automation and AI

- Built PHP cron automation for patient messaging that sends by location and time zone within an 8 AM–8 PM window, replacing manual sends.
- Built an AI ticketing tool where Claude (LLM API) structures business users' requests and MCP submits them to Jira, removing manual triage.
- Implemented RAG over the production codebase by chunking and embedding source code into a vector database for context-aware LLM responses.
- Use AI coding tools (Claude Code, Cursor, GitHub Copilot, OpenAI Codex) for coding, debugging, and documentation, and review all AI-generated code for logic, security, and performance issues before accepting, modifying, or rejecting it.

Performance and production support

- Troubleshoot and fix production defects and data inconsistencies across PHP, MySQL, JavaScript, and server environments using logs, SQL analysis, Git, and Jira tickets.
- Found the root cause of a slow production page that was hitting Cloudflare timeouts: a PHP $O(n^2)$ prescription-to-insurance comparison and repeated database queries. Replaced them with a hash-based lookup and removed the repeated queries, improving performance by about 60% and ending the timeouts.
- Rewrote a SQL date-range/HAVING query so filtering could use indexes, improving execution time by about 15%.
- Monitor Amazon EC2 instance health and application logs with AWS CloudWatch and use AWS CodePipeline to diagnose and fix deployment issues.

Application development and engineering practice

- Gather requirements from technical and non-technical stakeholders, then design, develop, test, and deploy PHP/MySQL web applications, including CRUD and API functionality and integrations with external APIs, releasing through Git-based CI/CD across dev, staging, and production on AWS.
- Built an interactive D3.js dashboard to track key KPIs and redesigned a frontend into a responsive KPI-style dashboard.
- Write unit and integration tests, take part in code reviews and pull requests, and document processes and system changes.

Youngstown State University — Youngstown, OH
Student Programmer I — 09/2024 – 06/2025

- Built and maintained custom PL/SQL applications integrated with the university's Banner ERP in a higher education environment, used by university departments to extract and process data and generate reports, reducing manual lookups from hours to minutes.

- Monitored production batch jobs, analyzed application and database logs, diagnosed job failures, and implemented fixes to improve reliability.
- Worked against production Banner databases and deployed through controlled job-submission workflows, coordinating with data custodians.
- Gathered requirements from department staff, then tested and revised applications with them in iterative review sessions.
- Managed code with SVN and TortoiseGit and worked tickets through Agile and Jira with a manager setting priorities.

PROJECTS

Multithreaded HTTP Server | C++17, POSIX Sockets, Linux, AWS EC2, Make

- Built a lightweight HTTP server in C++17 using POSIX sockets to accept TCP connections, parse HTTP GET requests, and serve static files.
- Implemented a custom thread pool using worker threads, a shared task queue, mutexes, and condition variables to handle concurrent client connections.
- Deployed and compiled the server on Ubuntu Linux using AWS EC2, validating HTTP responses with curl requests and browser testing.
- Configured AWS EC2 security group rules to control inbound access to the server on port 8080.
- Created a Makefile to automate compilation with C++17, compiler warnings, and POSIX thread support.

Job Board Scraper and Notifier | Python, FastAPI, SQLite, cron

- Built a FastAPI REST API over data pulled from the Adzuna API (200+ listings daily), with filtered queries, pagination, and correct HTTP status handling (200, 404, 422), averaging under 10 ms response time.
- Added B-tree indexes on title, location, and posted date, reducing query latency by about 60% across 1,000+ postings. Ran collection on a 6-hour cron with Discord and email notifications.

URL Shortener | Java, Spring Boot

- Built a URL shortener application in Java with Spring Boot that creates and resolves short links.

AI Document Parsing System | AI/LLM, Claude, Cursor

- Built an AI-based document parsing system, using Claude and Cursor for AI-assisted development.

C++ LRU File Cache | C++

- Implemented an in-memory LRU cache using hash-based lookup and a linked list for constant-time access and eviction. Measured about 42 ms on cache misses versus 0.4 ms on hits.

Virtual Disk File System Handler | C++, Operating Systems, ext2

- Built a C++ utility that transfers files between a host and VirtualBox VDI images with ext2 filesystems, parsing VDI headers and superblocks and traversing inodes through nested directories.
- Implemented bitmap-based block allocation and direct/indirect block pointer management, with metadata error-checking to validate filesystem consistency and prevent corruption.

EDUCATION

Youngstown State University Youngstown, OH
Bachelor of Science in Computer Science 12/2025
GPA 3.72/4.0 | Dean's List, President's List | Honors College Distinction | Minor: Mathematics | Certification: Data Analytics

TECHNICAL SKILLS

Languages: PHP, JavaScript, Python, Java, C++ (11/14/17), SQL, PL/SQL
Web & APIs: full-stack web applications, building and consuming REST APIs, third-party API integration, FastAPI, Java Spring Boot, HTML, CSS, jQuery, Bootstrap, pagination and error handling
Databases: MySQL, MSSQL, SQLite, MongoDB, PL/SQL; table and view design, index usage, query optimization
Cloud & DevOps: AWS (EC2, S3, CloudWatch, CodePipeline, SQS, SNS), GitHub Actions, CI/CD (dev, staging, production), Docker, cron-based automation, Linux, Bash
Enterprise Systems: Banner ERP, EMR integration, payment gateway integration, batch job monitoring, production support
Applied AI: Claude API, MCP, RAG, vector embeddings (Chroma), AI-assisted development (Claude Code, Cursor, GitHub Copilot, OpenAI Codex), reviewing AI-generated code
Data & Visualization: D3.js, Chart.js, dashboards, KPIs, data analysis, Tableau, R
Practices & Tools: Git, GitHub, pull requests, code reviews, SVN, TortoiseGit, Jira, Agile, unit testing, integration testing, debugging, troubleshooting, root-cause analysis, log analysis, technical documentation, Windows, Linux